

Section 10.1: Introduction**Goal:**

When we conduct a hypothesis test, we have a _____ and _____ hypothesis.

- **Null hypothesis (H_0):**

- **Alternative Hypothesis (H_a):**

Example 1. In the United States judicial system, a person is innocent until proven guilty. State the null and alternative hypotheses.

Since the null and alternative hypotheses are contradictory, we use _____ to determine if there is (or is not) enough evidence to conclude H_a is true.

Formally, there are two decisions we can make:

1.

2.

Section 10.2: Elements of a Statistical Test

Necessary Components of a Hypothesis Test (HT):

1. Null and alternative hypotheses
2. Assumptions/conditions
 -
3. Test statistic (TS)
 -
4. Rejection region (RR) or p-value
 - RR:
 - p-value:
5. Decision
6. Conclusion in context of the problem

Sometimes, we might make the wrong decision...

		Decision	
		Reject H_0	Fail to Reject H_0
Reality	H_0 is true		
	H_0 is false		

Probabilities of Errors:

- $P(\text{Type I error}) =$
 - referred to as
 - denoted by
- $P(\text{Type II error}) =$
 - denoted by

Fact: α and β are _____ related to one another.

- i.e., as α _____, β _____, and vice versa

Example 1 cont'd. Define α and β in context of the judicial system.

Example 2. It is claimed that the average height of all American males is 69 inches. We believe the average is higher than this. Suppose we obtain a random sample of 10 American males from a Normal population with variance 9, and we choose to use the rejection region $\{\bar{Y} \geq 71\}$.

(a) What are the null and alternative hypotheses?

(b) Calculate α .

(c) Calculate β if the true population mean height is 70 inches.

(d) Calculate β if the true population mean height is 72 inches.

Question. How can we improve our test?

Exercise. Repeat Example 2, but instead use a rejection region of $\{\bar{Y} \geq 70.5\}$.

(a) Calculate α .

(b) Calculate β if the true population mean height is 70 inches.

(c) Calculate β if the true population mean height is 72 inches.

Summary of Results:

	Rejection Region	
	$\{\bar{Y} \geq 71\}$	$\{\bar{Y} \geq 70.5\}$
α		
β ($\mu = 70$)		
β ($\mu = 72$)		

Section 10.3: Common Large-Sample Tests

Recall. In section 8.6 (*Large-Sample Confidence Intervals*), we gave the following table of common point estimators, along with their expected values and standard errors.

Table 8.1 Expected values and standard errors of some common point estimators

Target Parameter θ	Sample Size(s)	Point Estimator $\hat{\theta}$	$E(\hat{\theta})$	Standard Error $\sigma_{\hat{\theta}}$
μ	n	$\bar{Y}$	μ	$\frac{\sigma}{\sqrt{n}}$
p	n	$\hat{p} = \frac{Y}{n}$	p	$\sqrt{\frac{pq}{n}}$
$\mu_1 - \mu_2$	n_1 and n_2	$\bar{Y}_1 - \bar{Y}_2$	$\mu_1 - \mu_2$	$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}^{*\dagger}$
$p_1 - p_2$	n_1 and n_2	$\hat{p}_1 - \hat{p}_2$	$p_1 - p_2$	$\sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}^{\dagger}$

* σ_1^2 and σ_2^2 are the variances of populations 1 and 2, respectively.

$\dagger$ The two samples are assumed to be independent.

Recall. As a result of the CLT, all point estimators in Table 8.1 are approximately Normal *if the sample sizes are large enough*.

- This implies $Z = \frac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}}$ is approximately _____.

Def. For any of the target parameters in Table 8.1, we can perform an **upper-tailed test**; i.e.,

where θ_0 is a specific value of θ .

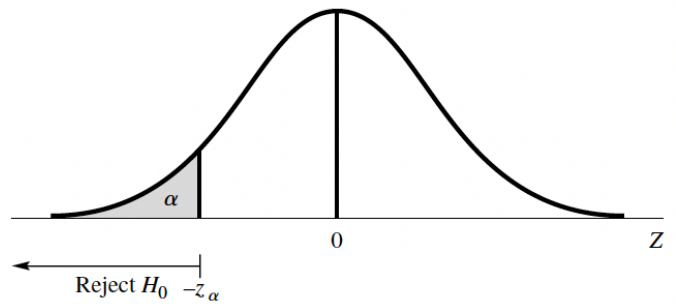
Remark. Large values of $\hat{\theta}$ favor _____.
Thus, a test statistic and rejection region can be written as follows:

Since larger values of $\hat{\theta}$ lead to larger values of Z , we can also use the following:

Question. What is the value of k_Z ?

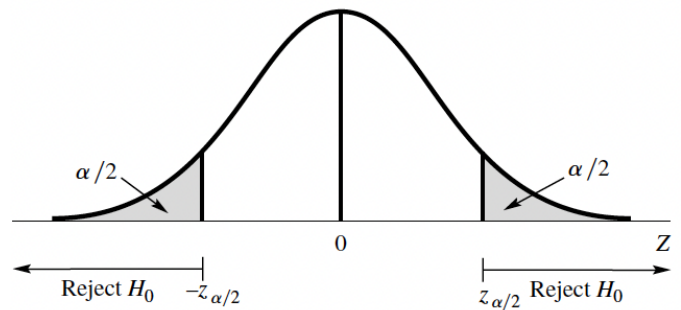
Example 3. A sociologist claims that the average number of people per family unit in a large subdivision is higher than the national average of 4.70. She randomly samples 100 families, obtaining a mean of 4.80 with a standard deviation of 0.80. Test her claim using $\alpha = 0.10$.

Def. A **lower-tailed test** is performed analogously:



Example 4. A real estate agent claims that less than 75% of retired couples prefer living in an apartment to owning their own home. Of 200 randomly sampled retired couples, 116 prefer living in an apartment. Test the agent's claim using $\alpha = 0.05$.

Def. For a **two-tailed test**, we reject H_0 if $\hat{\theta}$ is either much smaller or much larger than θ_0 . Thus, we have the following:



Example 5. A study was conducted to compare the reaction times of men and women to a stimulus. Independent random samples of 50 men and 50 women were taken. The men had a mean reaction time of 3.6 seconds and standard deviation of 0.42 seconds. The women had a mean reaction time of 3.8 seconds and standard deviation of 0.37 seconds. Do the data present sufficient evidence to suggest a difference between the true mean reaction times for men and women? Use $\alpha = 0.05$.

Summary:**Large-Sample α -Level Hypothesis Tests**

$$H_0 : \theta = \theta_0.$$

$$H_a : \begin{cases} \theta > \theta_0 & \text{(upper-tail alternative).} \\ \theta < \theta_0 & \text{(lower-tail alternative).} \\ \theta \neq \theta_0 & \text{(two-tailed alternative).} \end{cases}$$

$$\text{Test statistic: } Z = \frac{\hat{\theta} - \theta_0}{\sigma_{\hat{\theta}}}.$$

$$\text{Rejection region: } \begin{cases} \{Z > z_{\alpha}\} & \text{(upper-tail RR).} \\ \{Z < -z_{\alpha}\} & \text{(lower-tail RR).} \\ \{|Z| > z_{\alpha/2}\} & \text{(two-tailed RR).} \end{cases}$$

Section 10.4: Calculating Type II Error Probabilities and Finding Sample Size for Z Tests

Example 6. Suppose the sociologist in Example 3 has a specific alternative hypothesis in mind—that $\mu = 5$. Calculate β for this test.

Example 7. Suppose the real estate agent in Example 4 has a specific alternative hypothesis in mind—that $p = 0.42$. Calculate β for this test.

Question. How can we make β smaller?

Note: In the subsequent example and remarks, we assume the conditions of section 10.3 are satisfied (i.e., sample sizes are large enough).

Example 8. Suppose we want to test $H_0 : p = p_0$ vs. $H_a : p > p_0$. How large should the sample size be for a pre-specified α and β (where β is evaluated when $p = p_a$ and $p_a > p_0$)?

Remark. You get the same sample size formula for a lower-tailed test for a population proportion.

Remark. For a one-tailed test for a population mean, the sample size formula is

$$n = \frac{(z_\alpha + z_\beta)^2 \sigma^2}{(\mu_a - \mu_0)^2}.$$

Section 10.5: Relationships Between Hypothesis Tests and Confidence Intervals

So far...

- Section 8.6: The large-sample $100(1 - \alpha)\%$ CI for θ is
- Section 10.3: The large-sample α -level HT for $H_0 : \theta = \theta_0$ vs. $H_a : \theta \neq \theta_0$ uses the test statistic

and the rejection region

Question. What is the relationship between the CI and HT?

Remark. Thus, we fail to reject $H_0 : \theta = \theta_0$ in favor of $H_a : \theta \neq \theta_0$ if

The CI gives a range of plausible values for θ , so _____.

Section 10.6: Another Way to Report the Results of a Statistical Test: p-values

Def. The **p-value** (AKA **attained significance level**) is the

Remark. The smaller the p-value, _____.

Fact.

- If $\text{p-value} \leq \alpha$, we _____.
- If $\text{p-value} > \alpha$, we _____.

Example 9. Recall Example 3: “A sociologist claims that the average number of people per family unit in a large subdivision is higher than the national average of 4.70. She randomly samples 100 families, obtaining a mean of 4.80 with a standard deviation of 0.80. Test her claim using $\alpha = 0.10$.” Use the p-value approach to complete this problem.

Example 10. Recall Example 5: “A study was conducted to compare the reaction times of men and women to a stimulus. Independent random samples of 50 men and 50 women were taken. The men had a mean reaction time of 3.6 seconds and standard deviation of 0.42 seconds. The women had a mean reaction time of 3.8 seconds and standard deviation of 0.37 seconds. Do the data present sufficient evidence to suggest a difference between the true mean reaction times for men and women? Use $\alpha = 0.05$.” Use the p-value approach to complete this problem.

Section 10.8: Small-Sample Hypothesis Tests for μ and $\mu_1 - \mu_2$

Recall. In sections 8.8.1 and 8.8.2, we derived small-sample CIs for μ and $\mu_1 - \mu_2$. We had the following assumptions:

- $\underline{\mu}$: $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$.
- $\underline{\mu_1 - \mu_2}$: 2 independent random samples: $X_1, \dots, X_{n_1} \stackrel{i.i.d.}{\sim} N(\mu_1, \sigma_1)$ & $Y_1, \dots, Y_{n_2} \stackrel{i.i.d.}{\sim} N(\mu_2, \sigma_2)$.

Recall. Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$ and $T = \sqrt{n} \left(\frac{\bar{Y} - \mu}{S} \right)$. Then, $T \sim t_{n-1}$.

A Small-Sample Test for μ

Assumptions: $Y_1, Y_2, \dots, Y_n$ constitute a random sample from a normal distribution with $E(Y_i) = \mu$.

$H_0: \mu = \mu_0$.

$$H_a: \begin{cases} \mu > \mu_0 & \text{(upper-tail alternative).} \\ \mu < \mu_0 & \text{(lower-tail alternative).} \\ \mu \neq \mu_0 & \text{(two-tailed alternative).} \end{cases}$$

Test statistic: $T = \frac{\bar{Y} - \mu_0}{S/\sqrt{n}}$.

Rejection region: $\begin{cases} T > t_\alpha & \text{(upper-tail RR).} \\ T < -t_\alpha & \text{(lower-tail RR).} \\ |T| > t_{\alpha/2} & \text{(two-tailed RR).} \end{cases}$

Example 11. An organic chemist randomly sampled ten 4.85-g aniline specimens to produce acetanilide. The dry yield of acetanilide was measured for each specimen—the mean dry yield was 3.781 grams, and the standard deviation was 0.181 grams. Prior research shows that dry yields are normally distributed. The chemist wants to determine if there's evidence that the true mean dry yield of acetanilide differs from 4 grams. Test this claim using the p-value approach with $\alpha = 0.05$.

Small-Sample Tests for Comparing Two Population Means

Assumptions: Independent samples from normal distributions with $\sigma_1^2 = \sigma_2^2$.

$$H_0: \mu_1 - \mu_2 = D_0.$$

$$H_a: \begin{cases} \mu_1 - \mu_2 > D_0 & \text{(upper-tail alternative).} \\ \mu_1 - \mu_2 < D_0 & \text{(lower-tail alternative).} \\ \mu_1 - \mu_2 \neq D_0 & \text{(two-tailed alternative).} \end{cases}$$

$$\text{Test statistic: } T = \frac{\bar{X} - \bar{Y} - D_0}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}, \text{ where } S_p = \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}}.$$

$$\text{Rejection region: } \begin{cases} T > t_\alpha & \text{(upper-tail RR).} \\ T < -t_\alpha & \text{(lower-tail RR).} \\ |T| > t_{\alpha/2} & \text{(two-tailed RR).} \end{cases}$$

Section 10.10: Power of Tests and the Neyman-Pearson Lemma

Def. Suppose that W is the test statistic and RR is the rejection region for a hypothesis test involving the value of a parameter θ . Then, the **power** of the test, $\text{power}(\theta)$, is defined as

Scenario. Suppose we test the null hypothesis $H_0: \theta = \theta_0$ and that θ_a is a particular value of θ chosen from H_a .

(a) Evaluate $\text{power}(\theta_0)$.

(b) Evaluate $\text{power}(\theta_a)$.

Ideally, we would like $\alpha = \text{power}(\theta_0) = \underline{\hspace{1cm}}$ and $\text{power}(\theta_a) = \underline{\hspace{1cm}}$ for all θ_a in H_a , but this is impossible. Why?

Since we can't achieve the ideal situation, for a fixed sample size, we start with a small value of α and find the TS and RR that give the smallest possible value of β (i.e., the largest possible power).

Def. A **simple hypothesis** is one which

Any other hypothesis is a **composite hypothesis**.

Example 12. Suppose we have a Binomial probability distribution with $n = 10$ and unknown parameter p . Determine if the following hypotheses are simple or composite.

(a) $H_0 : p = 0.2$

(b) $H_a : p = 0.4$

(c) $H_a : p > 0.2$

Example 13. Consider the following scenarios.

(a) Suppose $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma = 1)$. Is $H_0 : \mu = 5$ a simple or composite hypothesis?

(b) Suppose $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$. Is $H_0 : \mu = 5$ a simple or composite hypothesis?

Theorem 10.1: Neyman-Pearson Lemma. Let $Y_1, \dots, Y_n$ be i.i.d. from some probability distribution with parameter θ . Suppose we want to test the simple null hypothesis $H_0 : \theta = \theta_0$ versus the simple alternative hypothesis $H_a : \theta = \theta_a$. For a given α , the test that maximizes the power at θ_a has a RR determined by

The value of k is chosen so that the test has the desired value for α . Such a test is a *most powerful* α -level test for H_0 versus H_a .

Example 14. Suppose Y represents a single observation from the probability density function given by

$$f(y|\theta) = \begin{cases} \theta y^{\theta-1}, & 0 < y < 1 \\ 0, & o.w. \end{cases}.$$

- (a) Find the most powerful test with significance level $\alpha = .05$ to test $H_0 : \theta = 2$ versus $H_a : \theta = 1$.

-
- (b) Calculate $\text{power}(\theta)$ for this test when $\theta = 1$.
- (c) Find the most powerful test with significance level $\alpha = .05$ to test $H_0 : \theta = 2$ versus $H_a : \theta = 3$.

Fact. We can sometimes use the Neyman-Pearson Lemma to find the **uniformly most powerful (UMP) test** of $H_0 : \theta = \theta_0$ vs. $H_a : \theta > \theta_0$ (or $H_0 : \theta = \theta_0$ vs. $H_a : \theta < \theta_0$).

- Our test is a UMP test of $H_0 : \theta = \theta_0$ vs. $H_a : \theta > \theta_0$ if the Neyman-Pearson test of $H_0 : \theta = \theta_0$ vs. $H_a : \theta = \theta_a$ is the same for all values of $\theta_a > \theta_0$.
- Our test is a UMP test of $H_0 : \theta = \theta_0$ vs. $H_a : \theta < \theta_0$ if the Neyman-Pearson test of $H_0 : \theta = \theta_0$ vs. $H_a : \theta = \theta_a$ is the same for all values of $\theta_a < \theta_0$.

Example 15. Suppose $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$, where σ is known. Find the α -level UMP test for testing $H_0 : \mu = \mu_0$ vs. $H_a : \mu > \mu_0$.

Exercise. Suppose $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$, where σ is known. Find the α -level UMP test for testing $H_0 : \mu = \mu_0$ vs. $H_a : \mu < \mu_0$.

Remark. From the previous example and exercise, we can infer that if we wish to test $H_0 : \mu = \mu_0$ vs. $H_a : \mu \neq \mu_0$, no single rejection region yields the most powerful test for all values of $\mu_a \neq \mu_0$.

Remark. The Neyman–Pearson Lemma is useless if we wish to test a hypothesis about a single parameter, θ , when the sampling distribution contains other unspecified parameters. Section 10.11 explains what to do in this situation.